

A dimension-order rate for the mean radius of every Orlicz ball

Candidate full answer to the centering question in arXiv:2011.07523

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Status. Candidate full affirmative solution, subject to human verification. For the uniform measure on an Orlicz ball, the normalized second moment differs from its Gibbs limit by $O(n^{-1})$. This is stronger than the $o(n^{-1/2})$ rate needed to transfer the source paper's complete moderate-deviation range from the limiting radius to the root-mean-square radius.

1 Source and exact question

The source is D. Alonso-Gutiérrez and J. Prochno, *Thin-shell concentration for random vectors in Orlicz balls via moderate deviations and Gibbs measures* [1], arXiv:2011.07523, published in *Journal of Functional Analysis* 282 (2022), 109291.

Let $M : \mathbb{R} \rightarrow [0, \infty)$ be even and convex, with $M(0) = 0$ and $M(x) > 0$ for $x \neq 0$, and put

$$B_M^n(nR) = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n M(x_i) \leq nR \right\}.$$

The source chooses $\alpha_* < 0$ so that the Gibbs random variable Z with density

$$p_M(x) = \frac{e^{\alpha_* M(x)}}{\int_{\mathbb{R}} e^{\alpha_* M(y)} dy}$$

satisfies $\mathbb{E}M(Z) = R$, and writes $L_Z^2 = \mathbb{E}Z^2$. Its Theorem B gives a sharp moderate-deviation upper estimate around L_Z when $M(x) = \Omega(x^2)$. In Remark 3.4, on printed and PDF page 13, the authors state that they do not know whether

$$\left(\mathbb{E} \frac{\|X_n\|_2^2}{n} \right)^{1/2} \longrightarrow L_Z$$

fast enough to replace L_Z by the root-mean-square radius in those estimates. Their next theorem performs the transfer under the stronger assumption $M(x) = \Omega(x^4)$. Figure 1 reproduces the full question and its immediate context.

2 Proof intuition

The uniform measure on the ball is an exact, one-sided exponential tilt of the product Gibbs measure. If

$$U = M(Z) - R, \quad S_n = U_1 + \cdots + U_n, \quad \beta = -\alpha_* > 0,$$

Remark 3.4. While thin-shell concentration inequalities are usually stated for the random variable $\frac{\|X_n\|_2}{\sqrt{n}}$ conveniently normalized, in the above result we obtained a thin-shell concentration inequality for the random variable $\frac{\|X_n\|_2^2}{nL_Z^2}$. An inequality for the random variable $\frac{\|X_n\|_2}{\sqrt{nL_Z}}$ is immediately obtained since for any $t > 0$, as $\frac{\|X_n\|_2}{\sqrt{nL_Z}} \geq 0$, we have that

$$\mathbb{P} \left[\left| \frac{\|X_n\|_2}{\sqrt{nL_Z}} - 1 \right| \geq t \right] = \mathbb{P} \left[\left| \frac{\|X_n\|_2^2}{nL_Z^2} - 1 \right| \geq t \left| \frac{\|X_n\|_2}{\sqrt{nL_Z}} + 1 \right| \right] \leq \mathbb{P} \left[\left| \frac{\|X_n\|_2^2}{nL_Z^2} - 1 \right| \geq t \right],$$

Notice that while Theorem B shows that in high dimensions if X_n is a random vector uniformly distributed on $B_M^n(nR)$, then $\frac{\|X_n\|_2}{\sqrt{n}}$ concentrates around L_Z , which is (as we will see) the asymptotic value of $\left(\mathbb{E} \left[\frac{\|X_n\|_2^2}{n} \right] \right)^{1/2}$, the thin-shell width conjecture is a conjecture on the concentration of $\frac{\|X_n\|_2}{\sqrt{n}}$ around $\left(\mathbb{E} \left[\frac{\|X_n\|_2^2}{n} \right] \right)^{1/2}$. We do not know if, in general, $\left(\mathbb{E} \left[\frac{\|X_n\|_2^2}{n} \right] \right)^{1/2}$ converges fast enough to L_Z so that estimates on the concentration around L_Z imply the same asymptotic estimates on the concentration around $\left(\mathbb{E} \left[\frac{\|X_n\|_2^2}{n} \right] \right)^{1/2}$. However, the next result shows that under some extra condition on the growth of the function M , we can transfer the concentration around L_Z to concentration around $\left(\mathbb{E} \left[\frac{\|X_n\|_2^2}{n} \right] \right)^{1/2}$, like in (2). Furthermore, in such case the estimate (1) is also improved for large values of t .

Theorem 3.5. *Let $M : \mathbb{R} \rightarrow [0, \infty)$ be an Orlicz function such that $M(x) = \Omega(x^4)$ as $x \rightarrow \infty$, $R > 0$, $n \in \mathbb{N}$, $\varphi_M : (-\infty, 0) \rightarrow \mathbb{R}$ the log-partition function with potential M and let $(Z_i)_{i \in \mathbb{N}}$ be a sequence of independent random variables with Gibbs density p_M , i.e.,*

$$\varphi_M(\alpha) = \log \int_{\mathbb{R}} e^{\alpha M(x)} dx \quad \text{and} \quad p_M(x) = \frac{e^{\alpha_* M(x)}}{\int_{\mathbb{R}} e^{\alpha_* M(x)} dx},$$

Figure 1: Remark 3.4 and the start of Theorem 3.5 on page 13 of [1].

then the Radon–Nikodym weight is $e^{\beta S_n} \mathbf{1}_{\{S_n \leq 0\}}$. Its mass is of order $n^{-1/2}$ and is concentrated in an $O(1)$ boundary layer immediately below zero.

Condition on the first coordinate. The conditional boundary-layer weight is a function $H_{n-1}(U_1)$. A third-order Edgeworth expansion for the sum of the remaining energies shows more than the naive fact that H_{n-1} is almost constant: after subtracting $H_{n-1}(0)$, its covariance with any centered first-coordinate observable having enough moments is $O(n^{-3/2})$. The mechanism is a second-difference cancellation. The normal term gains three powers of $n^{-1/2}$ because $\Phi''(0) = 0$; every Edgeworth correction has an explicit extra power of $n^{-1/2}$.

For the observable $Z_1^2 - L_Z^2$, division by the $n^{-1/2}$ normalizing mass therefore gives an $O(n^{-1})$ shift. Since $n^{-1} = o(t_n)$ whenever $t_n \sqrt{n} \rightarrow \infty$, changing centers does not alter the source moderate-deviation exponent or prefactor.

3 A boundary-layer lemma

We isolate the only probabilistic estimate not already present in the source.

Lemma 1. *Let U be centered, nondegenerate, satisfy Cramér’s condition*

$$\limsup_{|t| \rightarrow \infty} |\mathbb{E} e^{itU}| < 1,$$

and have moments of every order. Let $(U_i)_{i \geq 2}$ be independent copies, $T_m = U_2 + \dots + U_{m+1}$, and,

for a fixed $\beta > 0$, set

$$H_m(u) = \mathbb{E} \left[e^{\beta(T_m+u)} \mathbf{1}_{\{T_m+u \leq 0\}} \right].$$

If (U, W) is a coupled pair with $\mathbb{E}W = 0$ and $\mathbb{E}[|W|(1 + |U|^2)] < \infty$, then

$$\mathbb{E}[WH_m(U)] = O(m^{-3/2}).$$

Proof. Write $\sigma^2 = \text{Var } U$ and let F_m be the distribution function of T_m . Integration by parts gives, for every $u \in \mathbb{R}$,

$$H_m(u) = F_m(-u) - \beta \int_0^\infty e^{-\beta v} F_m(-u - v) dv. \quad (1)$$

By the classical Edgeworth expansion under Cramér's condition [2, Chapter 2],

$$F_m(y) = \Phi\left(\frac{y}{\sigma\sqrt{m}}\right) + \sum_{j=1}^3 m^{-j/2} q_j\left(\frac{y}{\sigma\sqrt{m}}\right) + r_m(y), \quad \sup_y |r_m(y)| = o(m^{-3/2}), \quad (2)$$

where each q_j is a polynomial times the standard normal density. All second derivatives q_j'' are therefore bounded.

For a twice differentiable function f , define

$$(\mathcal{B}_m f)(u) = f\left(-\frac{u}{\sigma\sqrt{m}}\right) - \beta \int_0^\infty e^{-\beta v} f\left(-\frac{u+v}{\sigma\sqrt{m}}\right) dv.$$

Because $\beta \int_0^\infty e^{-\beta v} dv = 1$, the difference $(\mathcal{B}_m f)(u) - (\mathcal{B}_m f)(0)$ is an exponentially averaged second difference. Hence, if $\|f''\|_\infty < \infty$,

$$|(\mathcal{B}_m f)(u) - (\mathcal{B}_m f)(0)| \leq \frac{C_f |u|}{m}. \quad (3)$$

For $f = \Phi$ one gains another factor $m^{-1/2}$: since $\Phi''(x) = -x\phi(x)$,

$$|(\mathcal{B}_m \Phi)(u) - (\mathcal{B}_m \Phi)(0)| \leq \frac{C|u|(1 + |u|)}{m^{3/2}}. \quad (4)$$

Indeed, before integration in v the mixed second difference is bounded by $C|u|v(|u| + v)m^{-3/2}$, and all exponential moments of v are finite.

Apply the linear operator in (1) to (2). Equations (3) and (4), together with the explicit factors $m^{-j/2}$, yield

$$H_m(u) - H_m(0) = O\left(\frac{|u|(1 + |u|)}{m^{3/2}}\right) + o(m^{-3/2}),$$

where the final remainder is uniform in u (its contribution is at most $4 \sup_y |r_m(y)|$). Since $\mathbb{E}W = 0$,

$$\mathbb{E}[WH_m(U)] = \mathbb{E}\{W[H_m(U) - H_m(0)]\} = O(m^{-3/2})$$

by the assumed mixed moment bound. □

4 The mean-radius rate and the transfer

Theorem 2. *Let M be any Orlicz function in the sense of [1], let $R > 0$, and let X_n be uniform on $B_M^n(nR)$. With Z , α_* , and L_Z as above,*

$$\frac{1}{n} \mathbb{E} \|X_n\|_2^2 = L_Z^2 + O(n^{-1}). \quad (5)$$

Consequently,

$$\left(\frac{1}{n} \mathbb{E} \|X_n\|_2^2 \right)^{1/2} = L_Z + O(n^{-1}).$$

If in addition $M(x) = \Omega(x^2)$ and $n^{-1/2} \ll t_n \ll 1$, then the source's Theorem B remains valid with the limiting second moment replaced by the exact second moment:

$$\mathbb{P} \left[\left| \frac{\|X_n\|_2^2}{\mathbb{E} \|X_n\|_2^2} - 1 \right| \geq t_n \right] \quad (6)$$

$$\leq |\alpha_*| \sqrt{2\pi n \varphi_M''(\alpha_*)} \exp \left\{ -\frac{t_n^2 n L_Z^4}{2 \text{Var}(Z^2)} (1 + o(1)) \right\} (1 + o(1)). \quad (7)$$

The same right-hand side bounds

$$\mathbb{P} \left[\left| \frac{\|X_n\|_2}{(\mathbb{E} \|X_n\|_2^2)^{1/2}} - 1 \right| \geq t_n \right].$$

Proof. Put $\beta = -\alpha_* > 0$, let $Z_1, \dots, Z_n$ be independent copies of Z , and define

$$U_i = M(Z_i) - R, \quad S_n = \sum_{i=1}^n U_i, \quad D_n = \mathbb{E}[e^{\beta S_n} \mathbf{1}_{\{S_n \leq 0\}}].$$

Changing from Lebesgue measure to the product Gibbs measure gives the exact identity

$$\mathbb{E} f((X_n)_1) = \frac{\mathbb{E}[f(Z_1) e^{\beta S_n} \mathbf{1}_{\{S_n \leq 0\}}]}{D_n} \quad (8)$$

for every integrable one-coordinate observable f . The sharp normalizer estimate used in the source (its equation (3)) is

$$D_n = \frac{1 + o(1)}{\beta \sqrt{2\pi n \text{Var}(M(Z))}}. \quad (9)$$

We verify the hypotheses of Lemma 1. Convexity and $M(x) > 0$ for $x \neq 0$ make M strictly increasing on $[0, \infty)$. Its inverse there is concave and absolutely continuous on compact intervals. It follows directly from

$$\mathbb{P}(M(Z) \leq y) = \frac{2 \int_0^{M^{-1}(y)} e^{-\beta M(x)} dx}{\int_{\mathbb{R}} e^{-\beta M(x)} dx}$$

that $M(Z)$ has a Lebesgue density. Thus its characteristic function tends to zero at infinity, so $U = M(Z) - R$ satisfies Cramér's condition. Also $\mathbb{E} e^{\eta M(Z)} < \infty$ for every $0 < \eta < \beta$, hence U has moments of every order and is nondegenerate.

Let $W = Z^2 - L_Z^2$. Every Orlicz function grows at least linearly at infinity, so Z and W have moments of every order; in particular $\mathbb{E}[|W|(1 + |U|^2)] < \infty$. Condition on (U_1, Z_1) in (8), set $m = n - 1$, and use the function H_m of Lemma 1. Symmetry of the ball gives

$$\begin{aligned} \frac{1}{n} \mathbb{E} \|X_n\|_2^2 - L_Z^2 &= \mathbb{E}(X_n)_1^2 - L_Z^2 \\ &= \frac{\mathbb{E}[(Z_1^2 - L_Z^2)H_{n-1}(U_1)]}{D_n}. \end{aligned}$$

The numerator is $O(n^{-3/2})$ by Lemma 1, while (9) is of order $n^{-1/2}$. This proves (5). Since $L_Z > 0$, the square-root assertion follows.

For the transfer, set $Q_n = \|X_n\|_2^2/n$ and $\mu_n = \mathbb{E}Q_n$. From (5), $|\mu_n - L_Z^2| = O(n^{-1}) = o(t_n)$. Therefore

$$\{|Q_n/\mu_n - 1| \geq t_n\} \subseteq \{|Q_n/L_Z^2 - 1| \geq s_n\}, \quad s_n = t_n(1 + o(1)).$$

The sequence s_n still satisfies $n^{-1/2} \ll s_n \ll 1$. Applying source Theorem B at s_n gives (6), including the same moderate-deviation exponent and prefactor. Finally, for $0 < t_n < 1$,

$$\left\{ \left| \sqrt{Q_n/\mu_n} - 1 \right| \geq t_n \right\} \subseteq \{|Q_n/\mu_n - 1| \geq t_n\},$$

which proves the norm version. $\square$

5 Verification notes, limitations, and novelty

1. **The main cancellation is genuinely second order.** Merely using a Berry–Esseen bound would leave an error as large as the $n^{-1/2}$ denominator. The third-order Edgeworth remainder and the mixed second difference in (1) are what produce the $n^{-3/2}$ centered numerator.
2. **No smoothness of M is used.** Convexity supplies a concave inverse and hence an absolutely continuous law for $M(Z)$. This is enough for the one-dimensional Cramér condition. Dense nondifferentiability of M causes no difficulty.
3. **The rate theorem is stronger than needed.** Equation (5) holds for every Orlicz function. The quadratic lower-growth condition enters only when invoking the source moderate-deviation theorem.
4. **Dependence on a standard theorem.** The proof relies on the classical third-order Edgeworth expansion under Cramér’s condition. A reviewer should verify the normalization and the uniform $o(m^{-3/2})$ remainder in (2); all its moment and Cramér hypotheses are checked explicitly above.
5. **Exact sanity checks.** For $M(x) = x^2$, direct integration in the Euclidean ball gives

$$n^{-1} \mathbb{E} \|X_n\|_2^2 = \frac{nR}{n+2}, \quad L_Z^2 = R.$$

For $M(x) = |x|$, the Dirichlet formula for the cross-polytope gives

$$n^{-1} \mathbb{E} \|X_n\|_2^2 = \frac{2n^2 R^2}{(n+1)(n+2)}, \quad L_Z^2 = 2R^2.$$

Both exact models agree with (5).

6. **Bounded novelty check.** The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:2011.07523 and the core centering terms. Web searches through 9 August 2026 used the exact question sentence, the title, L_Z , “expectation convergence rate”, and “Edgeworth Orlicz ball”. The closest later work found is Fröhlich–Prochno [3], which proves bivariate Edgeworth and quantitative CLT results for a smoother Orlicz class (C^2 away from a discrete set). It does not state (5) or the exact transfer above, and it explicitly notes that its smoothness convention is stronger than the general convex convention of [1]. Novelty is therefore plausible but not guaranteed.
7. **Human-review recommendation.** Prioritize Lemma 1, especially the use of the order-three Edgeworth remainder and the uniform treatment of the remainder after applying (1). Then check the exact Gibbs identity and the event inclusion used to transfer the source estimate.

References

- [1] D. Alonso-Gutiérrez and J. Prochno, *Thin-shell concentration for random vectors in Orlicz balls via moderate deviations and Gibbs measures*, J. Funct. Anal. 282 (2022), 109291; arXiv:2011.07523.
- [2] R. N. Bhattacharya and R. Ranga Rao, *Normal Approximation and Asymptotic Expansions*, SIAM Classics in Applied Mathematics 64, 2010 reprint of the 1976 edition, Chapter 2.
- [3] L. Fröhlich and J. Prochno, *Sharp concentration phenomena in high-dimensional Orlicz balls*, J. Funct. Anal. 290 (2026), 111322; arXiv:2407.15579.